

Sprint 1 Plan

Product: CircleHome

Scrum Master: Pravin Agrawal-Chung

Team: Camille Yabu, Antonio Wydler, Pravin Agrawal-Chung, Cameron Cox, Jacqueline Lopez-Carbajal

Sprint Completion Date: 4/22/2026

Revision Number: 1.0

Revision Date: 04/28/2026

Goal: Set up the full project infrastructure and deliver a working end-to-end foundation where users can create accounts, form or join a household, create tasks, view them in a shared real-time feed, and mark them complete.

Task Listing by User Story

(1.1) — "As a new user, I want to create an account with an email and password so that I can access my household securely."

- Set up Firebase Authentication in the Flutter project (2 hrs)
- Build sign-up screen UI (email, password, confirm password fields) (3 hrs)
- Build login screen UI (2 hrs)
- Connect sign-up and login forms to Firebase Auth (2 hrs)
- Add basic form validation and error messaging (1 hr)
- Handle auth state persistence so users stay logged in (1 hr)

Next grup

Total for 1.1: 11 hours

(1.6) — "As a user, I want a reliable way to store my login information, tasks, and household information"

- Initialize Flutter project with folder structure and routing (2 hrs)
- Add Firebase dependencies (firebase_core, cloud_firestore, firebase_auth) to pubspec.yaml and configure for iOS/Android (2 hrs)
- Design and document Firestore schema for users, households, and tasks collections (2 hrs)
- Create Firestore security rules for basic read/write access control (2 hrs)
- Set up shared state management solution (e.g. Provider or Riverpod) (2 hrs)

Total for 1.6: 10 hours

(1.2) — "As a new user, I want to create a household or join one using an invite code so that I can coordinate with the people I live with."

- Build household creation screen (name input, submit) (2 hrs)
- Generate a unique invite code on household creation and store in Firestore (2 hrs)
- Build join household screen (enter invite code) (2 hrs)
- Write Firestore logic to look up household by invite code and add user as member (2 hrs)
- Navigate user to home screen after household creation or join (1 hr)

Total for 1.2: 9 hours

(1.3) — "As a household member, I want to create a task with a title, category, due date, and assigned member so that responsibilities are clearly defined."

- Build task creation form UI (title, category dropdown, due date picker, assignee dropdown) (3 hrs)
- Populate assignee dropdown dynamically from household members in Firestore (2 hrs)
- Write Firestore logic to save new task document to tasks collection (2 hrs)
- Add input validation (required fields, valid date) (1 hr)

Total for 1.3: 8 hours

(1.4) — "As a household member, I want to view all household tasks in a shared feed sorted by due date so that I always know what needs to be done."

- Build task feed screen with a scrollable list of task cards (3 hrs)
- Display title, category, assignee, due date, and completion status on each card (2 hrs)
- Query Firestore for all tasks belonging to the user's household, sorted by due date (2 hrs)

Total for 1.4: 7 hours

(1.7) — "As a household member, I want the task feed to update in real time when a teammate adds or completes a task so that I do not have to refresh the app."

- Replace one-time Firestore query with a real-time snapshot listener on the tasks collection (2 hrs)

- Ensure UI rebuilds automatically when snapshot updates (1 hr)
- Test concurrent updates from two devices/sessions (1 hr)

Total for 1.7: 4 hours

(1.5) — "As a household member, I want to mark a task as complete so that the feed reflects current status and the assignee gets credit."

- Add a checkbox or "Mark Complete" button to each task card (1 hr)
- Write Firestore logic to update task document status field to "complete" (1 hr)
- Reflect completed status visually in the feed (e.g. strikethrough, greyed out) (1 hr)

Total for 1.5: 3 hours

Team Roles

(like Frontend Dev, Scrum Notes, Backend Dev, Tech Lead (Firebase/Firestore), Full-Stack Dev, QA)

Camille Yabu:

Antonio Wydler:

Pravin Agrawal-Chung:

Cameron Cox:

Jacqueline Lopez-Carbajal:

Initial Task Assignment

Camille Yabu:

Antonio Wydler:

Pravin Agrawal-Chung:

Cameron Cox:

Jacqueline Lopez-Carbajal:

A sprint burndown tracks remaining work (in hours) against the ideal pace across the sprint's working days (04/09/2026 to 04/22/2026). The team updates the Actual Remaining column daily

during standup. If the actual line rises above the ideal line, the team discusses scope reduction or impediment removal.

Sprint Hour Totals by User Story

User Story	Hours
1.1 -- Account creation (auth)	11
1.6 -- Firebase / Firestore infrastructure	10
1.2 -- Household creation and invite code	9
1.3 -- Task creation form	8
1.4 -- Shared task feed	7
1.7 -- Real-time feed updates	4
1.5 -- Mark task complete	3
Total Sprint Hours	52

Daily Burndown Tracker

Sprint Day	Date	Ideal Remaining (hrs)	Actual Remaining (hrs)
Day 0	04/09 (Wed)	52.0	
Day 1	04/10 (Thu)	46.8	
Day 2	04/11 (Fri)	41.6	
Day 3	04/14 (Mon)	36.4	
Day 4	04/15 (Tue)	31.2	
Day 5	04/16 (Wed)	26.0	
Day 6	04/17 (Thu)	20.8	
Day 7	04/18 (Fri)	15.6	
Day 8	04/21 (Mon)	10.4	

Day 9	04/22 (Tue)	5.2
-------	-------------	-----

Day 10	04/22 (Wed)	0.0
--------	-------------	-----

At the midpoint standup on 04/16, the Scrum Master should review the burndown gap and confirm all five user stories are in progress. If 1.5 or 1.7 have not been started, flag for potential descope. This is also the checkpoint to raise any Firebase/Firestore integration blockers before they cascade into 1.4 and 1.7.